Spectral Super-Resolution of Multispectral Images Using Spatial–Spectral Residual Attention Network

Xiangtao Zheng, *Member, IEEE*, Wenjing Chen, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*— The spectral super-resolution of multispectral image (MSI) refers to improving the spectral resolution of the MSI to obtain the hyperspectral image (HSI). Most recent works are based on the sparse representation to unfold the MSI into the 2-D matrix in advance for subsequent operations, which results in that the spatial information of MSI cannot be fully explored. In this article, a spatial–spectral residual attention network (SSRAN) is proposed to simultaneously explore the spatial and spectral information of MSI for reconstructing the HSI. The proposed SSRAN is composed of the feature extraction part, the nonlinear mapping part, and the reconstruction part. Firstly, the multispectral features of the input MSI are extracted in the feature extraction part. Second, in the nonlinear mapping part, the spatial–spectral residual blocks are proposed to explore spatial and spectral information of MSI for mapping the multispectral features to the hyperspectral features. Finally, in the reconstruction part, a 2-D convolution is used to reconstruct the HSI from the hyperspectral features. Also, a neighboring spectral attention module is specially designed to explicitly constrain the reconstructed HSI to maintain the correlation among neighboring spectral bands. The proposed SSRAN outperforms the state-of-the-art methods on both simulated and real databases.**

***Index Terms*— Attention, convolutional neural networks, hyperspectral image (HSI), multispectral image (MSI), spatial–spectral features, spectral super-resolution.**

I. Introduction

HYPERSPECTRAL imaging technique can capture the reflectance of the remote sensing scene across the electromagnetic spectrum [1]. The captured hyperspectral images (HSIs) have hundreds of narrow spectral bands [2], which are useful for distinguishing different materials. Therefore, HSIs have been widely exploited in various applications, such as detection [3] and land-cover recognition [4], [5]. However, the acquisition of HSIs is expensive and time-consuming due to the limitations of hardware devices, which restricts the applications for HSIs [6], [7]. An efficient way to obtain the HSI at low cost is to improve the spectral resolution of a multispectral image (MSI) under the same scene, which is called MSI spectral super-resolution[1] (MSISSR) [1].

Recently, many MSISSR methods [8]–[10] have been proposed to exploit the sparse representation to reconstruct the desired HSI from the MSI. These sparse representation-based methods assume that MSI and the corresponding HSI share the same sparse coefficients since the MSI and its corresponding HSI contain the same observed scene [1], [10]. These sparse representation-based methods first learn the MSI dictionary and the HSI dictionary from the paired MSI and HSI in the training set. Then, the learned MSI dictionary is used to estimate the sparse coefficients of the MSI to be spectrally super-resolved. Finally, the learned HSI dictionary and the estimated sparse coefficients are utilized to reconstruct the desired HSI. Furthermore, to improve the reconstruction performance, the low-rank prior and the sparse prior are often exploited in recent sparse representation-based methods [9], [10]. In fact, the spatially adjacent pixels in an MSI or an HSI are likely to have similar spectral signatures because spatially adjacent pixels usually contain the same type of land covers [11]. Therefore, in the MSISSR task, the spatial information of spatially adjacent pixels in the MSI may contribute to reconstructing the desired HSI [12]. However, recent sparse representation-based methods usually unfold the 3-D MSI into the 2-D matrix, which results in that the spatial information of the MSI cannot be fully explored [13].

The MSISSR task is a highly ill-posed problem [1] since hundreds of spectral bands need to be reconstructed from the MSI (only a few spectral bands) [10], [14]. A task very similar to the MSISSR task, namely the RGBSSR task, has been widely studied [15], [16]. The RGBSSR task refers to reconstructing the contiguous spectra of the HSI from the three-channel RGB image [17]. The RGBSSR task usually reconstructs only the visual spectra (400–700 nm) from the RGB image [16]. The RGBSSR task can be taken as the special case of the MSISSR task. Recently, various deep learning-based methods [18], [19] have been designed for the RGBSSR task by learning the RGB-to-HSI mapping. Deep learning-based methods have shown better performance than

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[1]In this article, the super-resolution refers to improving the spectral resolution of images.

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sparse representation-based methods in the RGBSSR task [20]. However, these deep learning-based methods are specially designed for the RGBSSR task and cannot fully explore the spectral signatures of MSI for the MSISSR task. Exploring both the spatial information and spectral signatures of MSI is beneficial for MSISSR [21].

In this article, a spatial–spectral residual attention network (SSRAN) is proposed to learn the MSI-to-HSI mapping. The proposed SSRAN consists of three parts: feature extraction, nonlinear mapping, and reconstruction. Firstly, SSRAN utilizes a 2-D convolution to extract the multispectral features from the input MSI [22]. Second, in the nonlinear mapping part of SSRAN, the spatial–spectral residual blocks map the multispectral features to the hyperspectral features by exploring the spatial information and spectral signatures of MSI. The proposed spatial–spectral residual block is composed of two branches: one is utilized to extract spatial features from spatially adjacent pixels, and the other is used to extract spectral features from spectral signatures. Finally, in the reconstruction part of SSRAN, a 2-D convolution with 1×1 kernels is utilized to reconstruct the HSI based on the hyperspectral features, and a neighboring spectral attention module is proposed to refine the reconstructed HSI. As HSIs are densely sampled along electromagnetic spectrum [23], [24], HSIs usually have the prior that neighboring spectral bands are correlated [25], [26]. To explicitly maintain this prior in the reconstructed HSI, the neighboring spectral attention module is utilized to generate the attention weight to couple each spectral band of the reconstructed HSI with its local neighboring spectral bands. In addition, the proposed neighboring spectral attention module can be further embedded into the spatial–spectral residual block to recalibrate spatial–spectral features, which is conducive to learning the MSI-to-HSI mapping. Experimental results on both simulated and real databases show the superior performance of SSRAN.

In this article, the main contributions are summarized as follows.

1) To reconstruct HSI from MSI, an end-to-end network is proposed to directly learn the MSI-to-HSI mapping. The proposed SSRAN is a lightweight network with high computational efficiency.
2) To simultaneously explore the spatial information and spectral signatures, a spatial–spectral residual block is proposed to extract spatial features and spectral features of MSI in parallel. The parallel extraction of features contributes to transmitting spatial features and spectral features to the deep layers of SSRAN, which can improve the reconstruction performance.
3) To explicitly maintain the neighboring spectral correlation prior in the reconstructed HSI, a neighboring spectral attention module is proposed to refine the reconstructed HSI. The proposed neighboring spectral attention module is a flexible network component, which can be easily embedded into the spatial–spectral residual block to recalibrate spatial–spectral features.

The remaining sections are structured as follows. Recent works related to SSR are introduced in Section II. Section III describes the proposed SSRAN. In Section IV, experimental results are presented. The conclusion is finally drawn in Section V.

II. Related Work

In recent years, SSR has gained lots of attention [10], [18] because SSR is a feasible and efficient way to generate the HSI [20]. According to different types of input images, the SSR task can be partitioned into two categories: RGBSSR and MSISSR. The RGBSSR task aims to reconstruct the HSI from the RGB image [19], and the MSISSR task aims to reconstruct the HSI from the MSI [9]. The image colorization task, similar to the RGBSSR and MSISSR tasks [27], [28], also focuses on enhancing the number of spectral bands of input images. However, the image colorization task aims to generate the RGB image from the gray image. Recent RGBSSR methods and MSISSR methods are reviewed in this section.

A. RGBSSR

As deep networks have achieved remarkable performance in diverse tasks [29], many methods build deep networks to tackle the task of reconstructing the HSI from the RGB image [19], [30]. Since the RGB image usually contains abundant spatial structure information of observed scenes, spatially adjacent pixels in the RGB image probably correspond to the same object. Moreover, spatially adjacent pixels within the same object in HSI are likely to share similar spectral signatures [20]. Therefore, recent deep learning-based methods [31], [32] usually focus on exploring the spatial information from local adjacent pixels in the RGB image for learning the RGB-to-HSI mapping. Zhang *et al.* [30] proposed to utilize multiple function-mixture blocks to explore different-sized receptive fields of the RGB image for learning the RGB-to-HSI mapping. Li *et al.* [33] proposed to integrate spectral response prior into an adaptive weighted attention network. Yan *et al.* [34] utilized a Unet-based network to exploit the prior category information of observed objects to improve the reconstruction performance. Zhao *et al.* [35] proposed a hierarchical regression network to exploit different-scaled features for RGBSSR. Li *et al.* [19] proposed a hybrid residual attentional network (RAN) to reconstruct an HSI from an RGB image, which consists of a 2-D RAN for extracting spatial features and a 3-D RAN for modeling interband correlations. Fu *et al.* [21] proposed a spectral–spatial convolutional neural network (SSCNN), including a spectral network for learning the spectral nonlinear mapping and a spatial network for exploiting the spatial similarity. In SSCNN, the spectral network and the spatial network are connected in series to learn the RGB-to-HSI mapping, where the spectral features of the spectral network cannot be directly transmitted to deep layers.

To learn the RGB-to-HSI mapping, the deep learning-based methods usually utilize the 2-D (or 3-D) convolution with 3×3 kernels to build the spatial feature extraction module for the RGB image [18], [20]. These deep learning-based RGBSSR methods [31], [32] only consider the spatial

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information, which ignores the spectral signatures. The 2-D (or 3-D) convolution is adept at extracting the spatial features of the MSI but is weak in extracting the spectral features of the MSI [14]. Therefore, these deep networks cannot explore the spectral signatures of multispectral pixels to learn the MSI-to-HSI mapping.

B. MSISSR

In the MSISSR task, most methods are based on the sparse representation [9], [10]. Gao *et al.* [9] first extracted the low-rank HSI dictionary and MSI dictionary from the paired MSI and HSI in the training set. Then, the extracted low-rank MSI dictionary is utilized to obtain the sparse coefficients of MSI. Finally, the desired HSI is reconstructed based on the extracted low-rank HSI dictionary and sparse coefficients. Yi *et al.* [10] proposed a spectral improvement constraint and a spatial preservation constraint based on the sparse representation. Fotiadou *et al.* [1] proposed a coupled dictionary learning framework to improve the spectral resolution. Han *et al.* [6] proposed to recover the lost spectral information of the MSI based on the spectral library. At first, the common spectral library is mapped to a specific spectral library by band matching. Then, the spectral dictionary and the sparse coefficient are extracted based on the specific spectral library. Recent MSISSR methods mainly build shallow learning models based on the sparse and low-rank priors, which results in the limited generalization ability [13].

Although the goals of the RGBSSR task and the MSISSR task are both to reconstruct the HSI, these two tasks have obvious differences. First, the inputs in the MSISSR task contain more spectral information than the inputs in the RGBSSR task [9]. Second, the reconstructed HSI in the RGBSSR task usually contains only dozens of spectral bands (e.g., 31) from 400 to 700 nm [16], whereas the reconstructed HSI in the MSISSR task often contains hundreds of spectral bands from 400 to 2500 nm [8], [10]. Therefore, directly applying the deep networks for the RGBSSR task to the MSISSR task may cause performance degradation [13]. Recently, Mei *et al.* [13] proposed a spatial–spectral joint super-resolution network (SSJSRN) to jointly improve both the spatial and spectral resolution of the MSI. SSJSRN is composed of a spatial super-resolution subnetwork and a spectral super-resolution subnetwork. This spectral super-resolution subnetwork is a plain network, where many 3-D convolutions are directly connected in series [13]. However, the spectral super-resolution subnetwork of SSJSRN cannot explicitly make the reconstructed HSI maintain the correlation of neighboring spectral bands.

III. Proposed Method

A. Overview

This article proposes an SSRAN to simultaneously explore the spatial information and spectral signatures of MSI for MSISSR. In the training stage, the proposed SSRAN directly learns an end-to-end MSI-to-HSI mapping from the paired MSI and HSI in the training set. In the testing stage, the desired HSI can be generated by directly feeding the MSI into the trained SSRAN. The overview of SSRAN is shown in Fig. 1. Inspired by the classic spatial super-resolution method [22], the overall framework of SSRAN is composed of three parts: feature extraction, nonlinear mapping, and reconstruction.

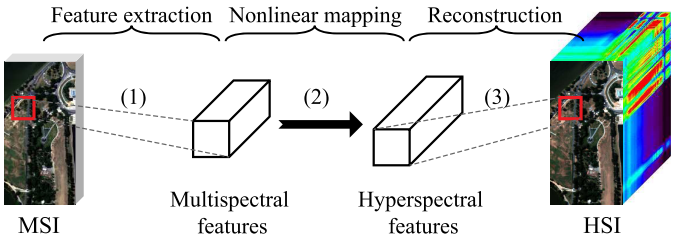


Fig. 1. Overview of the proposed SSRAN. SSRAN is composed of three parts. 1) feature extraction; 2) nonlinear mapping; and 3) reconstruction. The multispectral features of the MSI patch are first extracted in the feature extraction part and then are mapped into the hyperspectral features in the nonlinear mapping part. Finally, the HSI patch is reconstructed from the hyperspectral features in the reconstruction part.

1) *Feature Extraction:* This part takes the MSI patches as the input and converts the input from the image space to the feature space. The feature extraction part is utilized to extract the multispectral features of the input.
2) *Nonlinear Mapping:* This part aims to nonlinearly map the multispectral features of the input to the hyperspectral features. In this part, the spatial information and spectral signatures of the input are explored to obtain the hyperspectral features.
3) *Reconstruction:* This part is utilized to reconstruct the HSI patches based on the hyperspectral features obtained in the above nonlinear mapping part. The reconstruction part converts the hyperspectral features from the feature space to the image space.

The details of the feature extraction, nonlinear mapping, and reconstruction parts are introduced in the following subsections.

B. Feature Extraction

Let $X \in \mathbb{R}^{W\times H\times B_m}$ denote an MSI in the training set, where $W \times H$ represents the spatial size of the MSI X and B_m is the number of spectral bands in the MSI X. The ground truth HSI corresponding to the MSI X is denoted by $Y \in \mathbb{R}^{W\times H\times B_h}$, where B_h is the number of spectral bands in the ground truth HSI Y and $B_h \gg B_m$. Since MSISSR only focuses on improving the spectral resolution, X and Y have the same spatial size, contain the same observed scene, and share the same ground sampling distance [10]. A multispectral pixel $X(i, j) \in \mathbb{R}^{B_m}$ at the (i, j) position of X has a corresponding hyperspectral pixel $Y(i, j) \in \mathbb{R}^{B_h}$ in Y at the same spatial position. In the proposed SSRAN, the MSI-to-HSI mapping $M : \mathbb{R}^{B_m} \to \mathbb{R}^{B_h}$ is learned from the paired X and Y in the training set. The loss function is given as follows:

$$\text{Loss} = ||Y - M(X, \theta)||_F^2 \tag{1}$$

where θ represents all weight parameters of SSRAN and $||\cdot||_F$ is the Frobenius norm.

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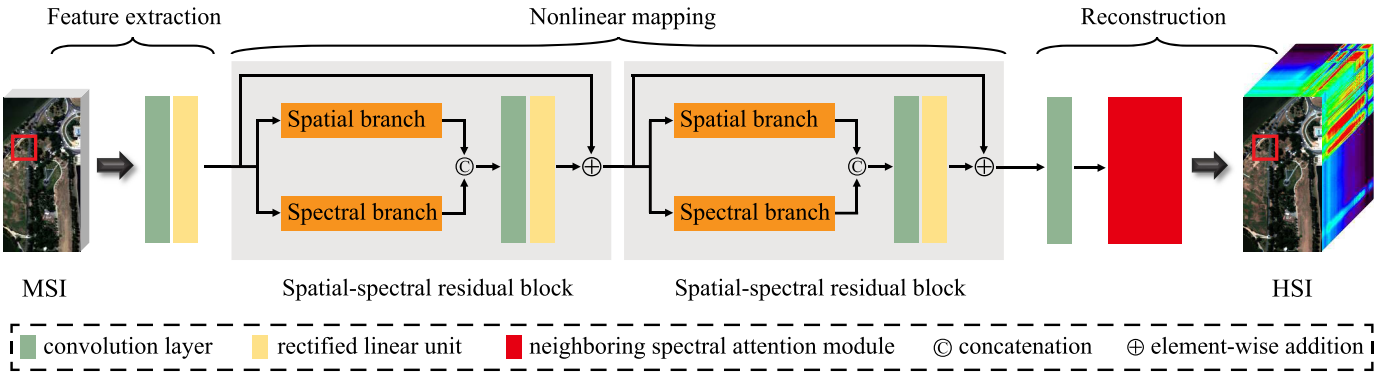


Fig. 2. Detailed structures of the proposed SSRAN. First, a convolution layer is utilized to extract the multispectral features from the cropped MSI patch. Then, several spatial–spectral residual blocks are utilized to obtain the hyperspectral features through the nonlinear mapping. Finally, a convolution layer and a neighboring spectral attention module are used to convert the obtained hyperspectral features to the HSI patch.

The detailed structures of the proposed SSRAN are shown in Fig. 2. To explore the spatial information and spectral signatures of local adjacent pixels, the MSI and the HSI are first cropped into plenty of paired MSI patches and HSI patches. Then, paired MSI patches and HSI patches are utilized to train the proposed SSRAN. Let $X_{\text{patch}} \in \mathbb{R}^{p\times p\times B_m}$ and $Y_{\text{patch}} \in \mathbb{R}^{p\times p\times B_h}$ represent a paired MSI patch and HSI patch cropped from X and Y, respectively, where $p \times p$ is the spatial size of patches.

The feature extraction part of SSRAN aims to convert the input MSI patches into the feature space. The structure of the feature extraction part is inspired by the spatial super-resolution method [22], where a 2-D convolution layer with a rectified linear unit (ReLU) is employed in the feature extraction part to extract features from the input MSI patch X_{patch}. In fact, it is unnecessary to use multiple convolutions in the feature extraction part because plentiful convolutions have been employed for obtaining the hyperspectral features in the nonlinear mapping part. Since the MSISSR task focuses on improving the spectral resolution, the spatial size of kernels in the 2-D convolution layer is set to 1×1 to extract the multispectral features $F_M \in \mathbb{R}^{p\times p\times C_m}$ from the MSI patch X_{patch} in the spectral dimension, where C_m is the number of feature channels in F_M.

C. Nonlinear Mapping

The nonlinear mapping part of SSRAN aims to obtain the hyperspectral features $F_H \in \mathbb{R}^{p\times p\times C_h}$ from the multispectral features F_M, where C_h is the number of feature channels in F_H. Since both spatial information and spectral signatures are helpful to learn the MSI-to-HSI mapping M [13], a spatial–spectral residual block is proposed to simultaneously explore the spatial information and spectral signatures of the MSI. The proposed spatial–spectral residual block is composed of a spatial branch and a spectral branch, where the spatial branch is utilized to extract the spatial features from F_M, and the spectral branch focuses on extracting the spectral features from F_M. Then, the spatial features and the spectral features are aggregated into the spatial–spectral features for generating the F_H in subsequent operations.

First, a spatial branch and a spectral branch are exploited to extract spatial features and spectral features, respectively. The spatial branch is composed of two 2-D convolutions with 3×3 kernels and two ReLUs, where 2-D convolutions with 3×3 kernels have high computational efficiency and have shown the excellent ability in the spatial feature extraction [20], [21]. Similar to the spatial branch, the spectral branch is simply composed of two 2-D convolutions with 1×1 kernels and two ReLUs. Since 1×1 kernels can make 2-D convolutions not introduce the spatial information from neighboring pixels [14], the 2-D convolutions with 1×1 kernels enable the spectral branch to extract spectral features only from the spectral dimension of MSI.

Second, a concatenation layer is employed to aggregate the spatial features from the spatial branch and the spectral features from the spectral branch in the feature channels. In addition, a 2-D convolution with 1×1 kernels and a ReLU are added following the concatenation layer to generate the spatial–spectral features. In previous methods [14], [21], the spectral branch and the spatial branch are usually connected in series to extract spatial–spectral features. However, in series connection, the spatial branch introduces the spatial information into the spectral features of the spectral branch to form the spatial–spectral features, which can result in that the spectral features of the spectral branch cannot be directly transmitted to deep layers of the deep network. In the proposed spatial–spectral residual block, the spatial branch and the spectral branch are connected in parallel to ensure that both spatial and spectral features can be directly used to generate the spatial–spectral features.

Finally, the residual connection [36] is employed in the proposed spatial–spectral residual block to facilitate convergence [31], [32]. In the proposed SSRAN, several spatial–spectral residual blocks are connected in series to form the nonlinear mapping part, which can increase the nonlinearity [22]. The spatial–spectral features of the last spatial–spectral residual block in the nonlinear mapping part are regarded as the hyperspectral features F_H.

D. Reconstruction

The reconstruction part of SSRAN aims to reconstruct the desired HSI from the hyperspectral features F_H. Similar to previous methods [33], [35], a 2-D convolution with B_h 1×1

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kernels is first exploited to convert the hyperspectral features F_H into the desired HSI patch $\bar{Y}_{\text{patch}} \in \mathbb{R}^{p\times p\times B_h}$, where B_h is the number of spectral bands in the HSI patch and $p \times p$ is the spatial size of the HSI patch. Different spectral bands of $\bar{Y}_{\text{patch}}$ are obtained by convolution using different 1×1 kernels in the 2-D convolution and the hyperspectral features F_H. However, since neighboring kernels in the 2-D convolution are not necessarily correlated [37], only using 2-D convolution to reconstruct the HSI patch from the hyperspectral features F_H cannot fully maintain the neighboring spectral correlation prior [23], [25] of the reconstructed HSI patch. The neighboring spectral correlation prior refers to that every spectral band of the HSI has high correlation with its local neighboring spectral bands [38], [39]. Maintaining the neighboring spectral correlation prior helps to improve the quality of the reconstructed HSI patch.

To further illustrate that 2-D convolution with 1×1 kernels cannot fully maintain the neighboring spectral correlation prior, the process of using 2-D convolution to reconstruct the HSI patch is introduced in detail. For simplicity, the spatial size of the input MSI patch is set to 1×1 for the introduction. Thus, the spatial size of the hyperspectral features F_H and the corresponding reconstructed HSI patch $\bar{Y}_{\text{patch}}$ is also 1×1. The shape of F_H is converted to $\mathbb{R}^{C_h}$ and the shape of $\bar{Y}_{\text{patch}}$ is converted to $\mathbb{R}^{B_h}$. The ith spectral band $\bar{Y}_{\text{patch}}(i)$ of $\bar{Y}_{\text{patch}}$ can be calculated as follows:

$$\bar{Y}_{\text{patch}}(i) = \sum_{j=1}^{C_h} W^i_{\text{conv}}(j) F_H(j) \tag{2}$$

where $W^i_{\text{conv}} \in \mathbb{R}^{C_h}$ represents the ith kernel in this 2-D convolution with 1×1 kernels. The spectral signature value of $\bar{Y}_{\text{patch}}(i)$ is determined by the weight parameter of W^i_{conv}. Similarly, the spectral signature values of $\bar{Y}_{\text{patch}}(i-1)$ and $\bar{Y}_{\text{patch}}(i+1)$, which are the local neighboring spectral bands of $\bar{Y}_{\text{patch}}(i)$, are determined by the weight parameters of W^{i-1}_{conv} and W^{i+1}_{conv}, respectively, However, the calculations of $\bar{Y}_{\text{patch}}(i-1)$, $\bar{Y}_{\text{patch}}(i)$, and $\bar{Y}_{\text{patch}}(i+1)$ are separated. Weight parameters of kernels W^{i-1}_{conv}, W^i_{conv} and W^{i+1}_{conv} are generally supervised by only the loss function without additional regularization in training, which cannot explicitly make kernels W^{i-1}_{conv}, W^i_{conv} and W^{i+1}_{conv} correlated. This may result in that the 2-D convolution with 1×1 kernels cannot explicitly maintain the neighboring spectral correlation prior in the reconstructed HSI patch $\bar{Y}_{\text{patch}}$ [10], [13], [40].

To explicitly make neighboring spectral bands of $\bar{Y}_{\text{patch}}$ correlated, a simple idea is to further couple each spectral band with its local neighboring spectral bands in $\bar{Y}_{\text{patch}}$. As shown in Fig. 3(a), a simple coupling operation is to calculate the average value of the spectral band with its k neighboring spectral bands, where k is an even number because neighboring bands usually exist in pairs. The coupling operation can be formulated as

$$\bar{Y}_{\text{coup}}(i) = \sum_{j\in\Omega_j} W_{\text{coup}}(j - i + \lfloor k/2+1 \rfloor)\bar{Y}_{\text{patch}}(j) \tag{3}$$

where $\bar{Y}_{\text{coup}} \in \mathbb{R}^{p\times p\times B_h}$ is the HSI patch refined by the coupling operation, $W_{\text{coup}} \in \mathbb{R}^{k+1}$ is the weight parameter of the coupling operation, $\lfloor k/2+1 \rfloor$ represents the largest integer not greater than $k/2+1$, and $\Omega_j = \{j \,|\, |j-i| \le \lfloor k/2 \rfloor, j \in \mathbb{N}^+\}$. In the coupling operation, the calculation of different spectral bands in $\bar{Y}_{\text{coup}}$ uses the same weight parameter W_{coup}, which can promote the correlation among neighboring spectral bands in $\bar{Y}_{\text{coup}}$. If each element of $W_{\text{coup}}(i)$ is equal to $1/(k+1)$, (3) means calculating the average value of local neighboring spectral bands. In particular, W_{coup} in the coupling operation can be embedded into the deep network for the end-to-end training to properly couple local neighboring spectral bands. In the deep learning framework, (3) can be implemented conveniently using the 1-D convolution with one kernel, where the size of the kernel is $k+1$.

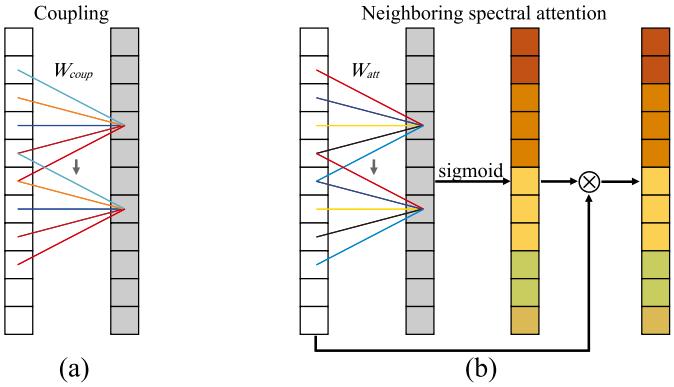


Fig. 3. (a) Coupling operation can be implemented using the 1-D convolution with one kernel. (b) Structures of the proposed neighboring spectral attention module. $\otimes$ represents the channel-wise multiplication.

However, the coupling operation mentioned above can only be added following the last 2-D convolution layer in the reconstruction part of SSRAN to refine the reconstructed HSI patch. Intermediate spatial–spectral features in the nonlinear mapping part cannot be fully exploited for learning the MSI-to-HSI mapping M. In this article, the coupling operation is further expanded into the neighboring spectral attention module to explore intermediate spatial–spectral features in the nonlinear mapping part. Detailed settings of the proposed neighboring spectral attention module are shown in Fig. 3(b). Take $\bar{Y}_{\text{patch}}$ as the input to introduce the neighboring spectral attention module. First, a 1-D convolution with a sigmoid activation is exploited to generate the attention weight $\alpha \in \mathbb{R}^C$ based on k neighboring spectral bands of $\bar{Y}_{\text{patch}}$. Then, the attention weight α is utilized to recalibrate the spectral bands of $\bar{Y}_{\text{patch}}$. The proposed neighboring spectral attention module can be formulated as follows:

$$\alpha(i) = \sigma\left(\sum_{j\in\Omega_j} W_{\text{att}}(j - i + \lfloor k/2+1 \rfloor)\bar{Y}_{\text{patch}}(j)\right) \tag{4}$$

$$\bar{Y}_{\text{att}} = \alpha \otimes \bar{Y}_{\text{patch}} \tag{5}$$

where $\bar{Y}_{\text{att}} \in \mathbb{R}^{p\times p\times C}$ represents the output of the neighboring spectral attention module, W_{att} represents the weight parameter of the neighboring spectral attention module, σ represents the sigmoid activation, and $\otimes$ represents the channel-wise multiplication. When the spatial size $p \times p$ of X_{patch} and $\bar{Y}_{\text{patch}}$ is greater than 1×1, a global average pooling is

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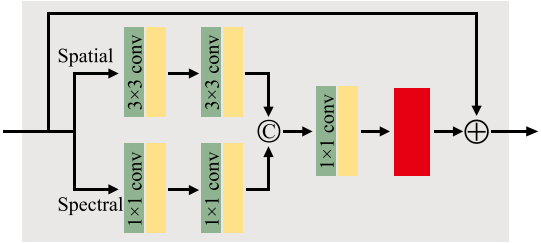


Fig. 4. Structures of the proposed spatial–spectral residual block combined with the neighboring spectral attention module. Green block: convolution layer. Yellow block: rectified linear unit. Red block: proposed neighboring spectral attention module.

added before the 1-D convolution in the neighboring spectral attention module for calculating the attention weight α. When the neighboring spectral attention module is added following the 2-D convolution layer in the reconstruction part instead of the above-mentioned coupling operation, $\bar{Y}_{\text{att}}$ is the final refined reconstructed HSI patch of the proposed SSRAN.

In addition, the neighboring spectral attention module can be embedded into the proposed spatial–spectral residual block to explore intermediate spatial–spectral features in the nonlinear mapping part to refine the hyperspectral features F_H. As shown in Fig. 4, inspired by the literature [37], [41], the neighboring spectral attention module is added before the residual connection of the spatial–spectral residual block to recalibrate the intermediate spatial–spectral features.

Similar to previous channel attention modules [37], [42], the proposed neighboring spectral attention module also focuses on generating the attention weights for recalibrating feature channels. Different from previous channel attention modules [37] that the fully connected layer is used to generate the attention weight, the proposed neighboring spectral attention module employs the 1-D convolution layer to generated the attention weight. In the proposed neighboring spectral attention module, using 1-D convolution layer has two advantages. One advantage is that 1-D convolution can facilitate neighboring spectral bands correlated. The other advantage is that the parameter number of 1-D convolution is far less than that of the fully connected layer, which avoids the introduction of too many weight parameters [41].

IV. Experiments

A. Databases

Due to various aerial and astronautic imaging sensors, different remote sensing databases usually have different ground sampling distances. To fully evaluate the performance of SSRAN, the Washington DC Mall, Pavia University, and Paris databases that have different ground sampling distances are employed to conduct experiments.

First, the Washington DC Mall (WDCM)[2] and Pavia University[3] databases are employed as the simulated databases. The WDCM database contains an HSI with 191 bands that is captured by the HYDICE sensor. The ground sampling distance of the WDCM database is 3.2 m. An HSI with 240 × 240 pixels is cropped from the WDCM database to conduct experiments. Following Wald's protocol [9], [10], a spectral response function that corresponds to Landsat TM bands 1–5 and 7 [43] (i.e., the 450–520, 520–600, 630–690, 760–900, 1550–1750 and 2080–2350 nm domains) is utilized to spectrally downsample the cropped HSI to generate its corresponding MSI. The Pavia University database contains an HSI with 103 bands that is captured by the ROSIS sensor. The ground sampling distance of the Pavia University database is 1.3 m. An HSI with 200 × 200 pixels is cropped from the Pavia University database. Following Wald's protocol, a spectral response function that corresponds to Landsat TM bands 1–4 [43] (i.e., the 445–516, 506–595, 632–698, and 757–853 nm domains) is utilized to spectrally downsample the cropped HSI to obtain its corresponding MSI.

Then, the Paris[4] [44] database that is a real database is further employed to evaluate the performance of SSRAN. The ground sampling distance of the Paris database is 30 m. The Paris database includes an HSI captured by the Hyperion[5] instrument and an MSI captured by the ALI[6] instrument. In practice, there is also a spectral response function that can be used to generate the MSI captured by the ALI instrument from the HSI captured by the Hyperion instrument [44]. Both HSI and MSI contain 72 × 72 pixels and have the same ground sampling distance. HSI contains 128 spectral bands and MSI has nine spectral bands.

As mentioned above, each database contains an HSI and an MSI. Similar to the literature [9], the subregions of HSI and MSI that are overlapped are selected as the training set on these databases. On the WDCM database, the training set contains 240 × 120 pixels. On the Pavia University database, the training set contains 200 × 100 pixels. On the Paris database, the training set contains 72 × 36 pixels. The rest regions of MSI on the databases are the MSI to be spectrally super-resolved. The rest regions of HSI on the databases are the ground truth. The false-color images of the training set, the MSI to be spectrally super-resolved, and the ground truth are shown in Fig. 5.

B. Evaluation Metrics

The quantitative results of all methods are evaluated with five widely used metrics, including root mean squared error (RMSE), spectral angle mapper (SAM), universal image quality index (UIQI), erreur relative globale adimensionnelle de synthèse (ERGAS), and peak signal-to-noise ratio (PSNR). Similar to the literature [45], the quantitative results are calculated based on the 8-bit images. In addition, the visual false-color images and error maps of the reconstructed HSIs are also given to qualitatively show the performance of SSRAN. Error maps represent the sum of absolute values between the ground truth and the reconstructed HSI in each spectral band.

[2]https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

[3]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes#Pavia_University_scene

[4]https://github.com/alfaiate/HySure

[5]http://eo1.usgs.gov/sensors/hyperioncoverage

[6]http://eo1.usgs.gov/sensors/ali

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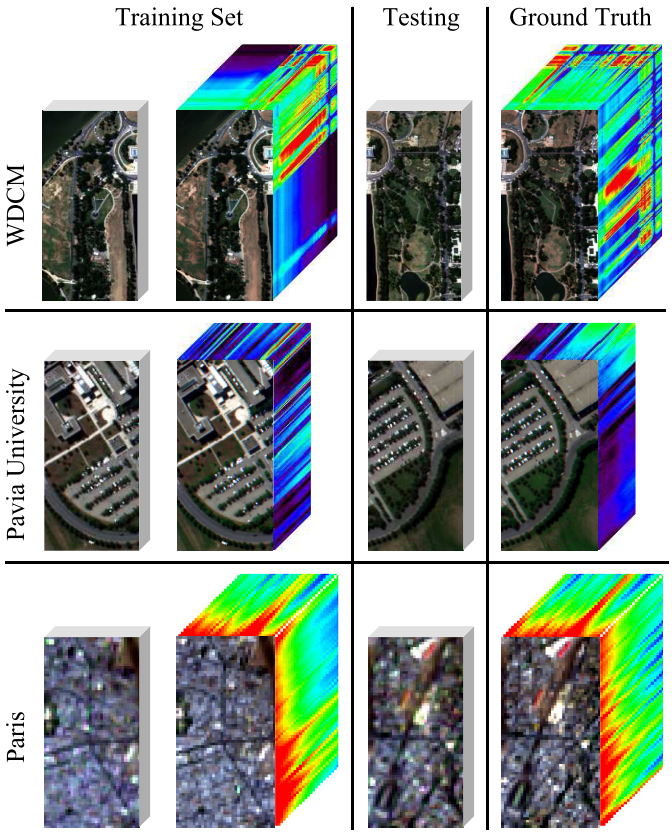


Fig. 5. False-color images of the training set, the MSI to be spectrally super-resolved, and the ground truth on the WDCM (R: 50, G: 30, and B: 20), Pavia University (R: 60, G: 30, and B: 10), and Paris (R: 24, G: 14, and B: 3) databases.

C. Detailed Settings of SSRAN

This section explores the impacts of different settings on the performance of SSRAN on the WDCM database, including the number of spatial–spectral residual blocks in the nonlinear mapping part and the number of neighboring spectral bands of the neighboring spectral attention module in the reconstruction part. The kernel number of the 2-D convolution layer in the reconstruction part is set to be the number of bands of the reconstructed HSI. In SSRAN, the kernel number of other convolution layers is set to 64. In addition, the performance of the proposed neighboring spectral attention module in the reconstruction part is also studied in this section. The spatial size of HSI and MSI patches is simply set to 4×4 for experiments since an observed object usually contains only a few pixels in the remote sensing HSIs. Details are given in the following.

1) Number of Spatial–Spectral Residual Blocks: The nonlinear mapping part of SSRAN is composed of several spatial–spectral residual blocks to obtain the hyperspectral features F_H from the multispectral features F_M. In the nonlinear mapping part, multiple spatial–spectral residual blocks can increase the nonlinearity but also introduce too many weight parameters. The number of spatial–spectral residual blocks is set to 1, 2, 3, 4, and 5 to conduct experiments. The number of neighboring spectral bands in the neighboring spectral attention module is initially set to 5 in experiments. Table I reports the SAM and RMSE of SSRAN. The proposed SSRAN with three spatial–spectral residual blocks obtains the best performance. Thus, the number of spatial–spectral residual blocks in the nonlinear mapping part is set to 3 in the following experiments.

TABLE I

RMSE AND SAM OF SSRAN WITH DIFFERENT NUMBERS OF SPATIAL–SPECTRAL RESIDUAL BLOCKS ON THE WDCM DATABASE

Number of Blocks	1	2	3	4	5
RMSE	0.858	0.909	**0.852**	0.859	0.872
SAM	0.959	0.952	**0.942**	0.964	0.954

TABLE II

RMSE AND SAM OF SSRAN WITH DIFFERENT NUMBERS OF NEIGHBORING SPECTRAL BANDS ON THE WDCM DATABASE

Number of Bands	1	3	5	7	9
RMSE	0.865	0.861	**0.852**	0.868	0.865
SAM	0.950	0.950	**0.942**	0.961	0.958

2) Number of Neighboring Spectral Bands: In the reconstruction part of SSRAN, the neighboring spectral attention module generates the attention weights based on its local neighboring spectral bands to facilitate the reconstructed HSI to maintain the neighboring spectral correlation prior. The number of neighboring spectral bands is an important parameter that affects the quality of the reconstructed HSI. Too many neighboring spectral bands are used to generate the attention weights, which may cause the generated hyperspectral signatures to be too smooth, and too few can cause the generated hyperspectral signatures to not maintain the neighboring spectral correlation prior. Table II reports the SAM and RMSE of SSRAN when this parameter is 1, 3, 5, 7, and 9. When the number of neighboring spectral bands is 5, SSRAN obtains the best performance on the WDCM database. Therefore, the number of neighboring feature channels in the neighboring spectral attention module is set to 5 in following experiments.

3) Analysis of the Neighboring Spectral Attention Module: In the reconstruction part of SSRAN, two coupling operations (the average value and the 1-D convolution) and a neighboring spectral attention module are proposed to explicitly maintain the neighboring spectral correlation prior of the reconstructed HSI. Since these coupling operations only focus on improving the correlation between neighboring spectral bands in the spectral dimension, a spectral residual attention network (SRAN) is built to verify the effectiveness of these coupling operations in the spectral dimension. SRAN is the proposed SSRAN removed the spatial branch. SRAN directly reconstructs hyperspectral pixels from multispectral pixels without introducing the spatial information. In experiments, the spatial size of network inputs is set to 1×1. To show the effectiveness of coupling local neighboring spectral bands, a 2-D convolution with B_h 1×1 kernels that can couple all spectral bands is employed as a comparison. In addition, to demonstrate the

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TABLE III
RMSE AND SAM OF DIFFERENT COUPLING OPERATIONS ON THE WDCM DATABASE

Methods	Basis	AVE	1-D Conv	2-D Conv	SE	SRAN
RMSE	0.904	0.896	0.893	0.895	0.893	**0.874**
SAM	0.991	0.989	0.977	0.989	0.973	**0.961**

effectiveness of the proposed neighboring spectral attention module, an existing squeeze-and-excitation channel attention module [37] is utilized to replace the neighboring spectral attention module in SRAN to conduct experiments.

1) *Basis* represents the proposed SSRAN removed the spatial branch in the nonlinear mapping part and the neighboring spectral attention module in the reconstruction part.
2) *AVE* represents that the average value operation is added following the 2-D convolution layer in the reconstruction part of the basis method.
3) *1-D Conv* represents that the 1-D convolution is added following the 2-D convolution layer in the reconstruction part of the basis method.
4) *2-D Conv* represents that the 2-D convolution is added following the 2-D convolution layer in the reconstruction part of the basis method.
5) *SRAN* represents that the neighboring spectral attention module is added following the 2-D convolution layer in the reconstruction part of the basis method.
6) *SE* represents that the neighboring spectral attention module in the reconstruction part of SRAN is replaced by the existing squeeze-and-excitation channel attention module [37].

The RMSE and SAM of different methods on the WDCM database are shown in Table III. As shown in Table III, the coupling operations can effectively improve the performance of the basis method. Although the AVE method and the 2-D Conv method have similar performance, the AVE method does not introduce any weight parameters that need to be trained. Comparing the performance of the 1-D Conv method and the 2-D Conv method, it can be found that coupling local neighboring spectral bands can effectively improve the quality of the reconstructed HSI. The SRAN method outperforms the SE method since the proposed neighboring spectral attention module can explicitly maintain the neighbor spectral correlation prior of the reconstructed HSI.

D. Ablation Study

In the nonlinear mapping part of SSRAN, a spatial–spectral residual block that consists of a spatial branch and a spectral branch is proposed to explore the spatial information and spectral signatures of MSI. In the reconstruction part of SSRAN, a neighboring spectral attention module is proposed to refine the reconstructed HSI. In this section, several potential networks are built to study the effectiveness of the spatial–spectral residual block and the neighboring spectral attention module for MSISSR. The details of potential networks are as follows.

1) *Spe-Net* represents the SSRAN without the spatial branch and the neighboring spectral attention module.
2) *Spa-Net* represents the SSRAN without the spectral branch and the neighboring spectral attention module.
3) *Spa-Spe-Net* represents the SSRAN without the neighboring spectral attention module.
4) *Spe-Att-Net* represents the SSRAN without the spatial branch.
5) *Spa-Att-Net* represents the SSRAN without the spectral branch.
6) *SSRAN* (Spa-Spe-Att-Net) is the proposed method.

Quantitative results of these potential networks are listed in Tables IV–VI. The proposed SSRAN obtains better performance than Spa-Spe-Net on three databases, which demonstrates that the neighboring spectral attention module can effectively improve the quality of the reconstructed HSI. On the WDCM database, Spe-Att-Net performs better than Spa-Att-Net. However, Spa-Att-Net performs better than Spe-Att-Net on the Pavia University and Paris databases. This shows that spatial information and spectral signatures have different importance for MSISSR on different databases. On the WDCM and Pavia University databases, the ground sampling distances are small, and a pixel may contain only a few endmembers or even one endmember. The compositions of spectral signatures are relatively simple. As shown in Tables IV and V, Spa-Spe-Net that is composed of both the spatial branch and the spectral branch shows better performance than Spe-Net and Spa-Net on the WDCM and Paiva University databases. This can demonstrate that both spatial information and spectral signatures are effective for MSISSR. When the ground sampling distance is large, a pixel may contain a large number of endmembers, which can make the compositions of spectral signatures in MSI complicated. Therefore, it is challenging to explore the spectral signatures of MSI on the Paris database to reconstruct the HSI. As shown in Table VI, on the Paris database, the spectral features from spectral signatures can reduce the experimental results of SSRAN.

In practice, with the rapid development of remote sensing imaging technology, the ground sampling distances of the captured MSIs have been improved. There are few cases where the ground sampling distance of MSI is very low (e.g., 30 m). Therefore, the proposed SSRAN utilizes the spectral branch, the spatial branch, and the neighboring spectral attention module as basic components for MSISSR.

E. Comparison With Other Methods

To evaluate the performance of SSRAN, the WDCM, Pavia University, and Paris databases are utilized to conduct experiments. Four recent representative methods are employed as the compared methods, including pixel-wise copy (PWC) [9], joint sparse and low-rank learning[7] (J-SLoL) [9], spectral–spatial convolutional neural network (SSCNN) [21], and spatial–spectral joint super-resolution network (SSJSRN) [13]. PWC [9] is a baseline MSISSR method for comparison, which is simply based on the copying

[7] https://github.com/danfenghong/IEEE_TGRS_J-SLoL

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operation. To spectrally super-resolve a multispectral pixel, PWC first finds the most similar multispectral pixel from the training samples. Then, the hyperspectral pixel corresponding to the most similar multispectral pixel is directly copied as the reconstructed hyperspectral pixel. In this article, the bicubic interpolation is not employed as a baseline comparison method because interpolating hundreds of bands (HSI) from several bands (MSI) will cause greater distortion. J-SLoL [9] is a recent sparse representation-based method. SSCNN [21] is a deep learning-based method designed for RGBSSR. SSJSRN [13] is composed of a spatial super-resolution subnetwork used for spatial super-resolution of MSI and a spectral super-resolution subnetwork used for spectral super-resolution of MSI. In SSJSRN, only the spectral super-resolution subnetwork is utilized for experiments. In addition, Spe-Net, Spa-Net, Spa-Spe-Net, Spe-Att-Net, and Spa-Att-Net are also employed as compared methods.

The implementation details of all methods are given in the following. PWC and J-SLoL use the same hyperparameter settings as the original literature [9]. SSCNN, SSJSRN, Spe-Net, Spa-Net, Spa-Spe-Net, Spe-Att-Net, Spa-Att-Net, and the proposed SSRAN utilize the same training epoch (200), batch size (100), and Xavier weight initialization [46] for a fair comparison. The learning rates of SSCNN and SSJSRN are the same as the settings of the original literature. The learning rates of Spe-Net, Spa-Net, Spa-Spe-Net, Spe-Att-Net, Spa-Att-Net, and the proposed SSRAN are initially set to 0.01 and then divided by 10 every 50 epochs. Paired MSI and HSI patches with the spatial size 4×4 are first cropped from the training set for training, where the stride of the cropping operation is 1. Then, 4×4 MSI patches are cropped from the MSI as testing samples. For simplicity, the stride of the cropping operation in the testing phase is set to 4. Experimental results of different methods are as follows.

1) On the WDCM Database: First, the WDCM database is utilized to conduct experiments. The ground sample distance of the WDCM database is 3.2 m. On the WDCM database, a pixel in MSI and HSI contains only a few endmembers or even one endmember. The compositions of spectral signatures in a pixel on the WDCM database are relatively simple, which makes it easy to learn the optimal MSI-HSI mapping. The quantitative results of different methods on the WDCM database are reported in Table IV. PWC shows poor performance because PWC directly copies the hyperspectral pixels of the training samples into the reconstructed HSI without any changes. In practice, even for the same category of land covers, the spectral signatures of their corresponding hyperspectral pixels will vary with the changes in the illumination and weather conditions [47], [48]. SSCNN and SSJSRN show better performance than J-SLoL in terms of RMSE and SAM. However, SSCNN and SSJSRN perform worse than J-SLoL in terms of UIQI, ERGAS, and PSNR. Different from SSCNN and SSJSRN, the proposed SSRAN aims to simultaneously explore the spatial information and spectral signatures of MSI. In addition, a neighboring spectral attention module is designed for MSISSR. The proposed SSRAN outperforms compared methods in terms of RMSE, SAM, UIQI, ERGAS, and PSNR.

TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE WDCM DATABASE

Methods	RMSE	SAM	UIQI	ERGAS	PSNR
PWC [9]	1.462	1.465	0.977	10.696	34.184
J-SLoL [9]	1.129	1.398	0.993	5.271	41.042
SSCNN [21]	1.002	1.139	0.990	7.522	38.315
SSJSRN [13]	0.916	1.067	0.982	10.592	38.946
Spe-Net	0.877	0.970	**0.994**	5.578	41.424
Spa-Net	0.900	1.042	**0.994**	6.408	40.593
Spa-Spe-Net	0.869	0.963	**0.994**	5.651	41.829
Spe-Att-Net	0.872	0.965	**0.994**	5.542	41.610
Spa-Att-Net	0.892	0.966	**0.994**	5.779	41.427
SSRAN	**0.858**	**0.946**	**0.994**	**5.189**	**41.938**

TABLE V
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATABASE

Methods	RMSE	SAM	UIQI	ERGAS	PSNR
PWC [9]	3.262	3.614	0.973	9.995	39.217
J-SLoL [9]	2.520	2.806	0.983	7.306	42.138
SSCNN [21]	2.242	2.507	0.984	7.129	43.178
SSJSRN [13]	2.180	2.408	**0.985**	6.810	43.021
Spe-Net	2.289	2.486	**0.985**	6.895	42.976
Spa-Net	2.204	2.446	**0.985**	6.854	43.231
Spa-Spe-Net	2.192	2.366	**0.985**	6.836	43.282
Spe-Att-Net	2.260	2.473	**0.985**	6.850	43.033
Spa-Att-Net	2.156	2.371	**0.985**	6.758	43.259
SSRAN	**2.154**	**2.341**	**0.985**	**6.709**	**43.379**

The visual results and error maps of different methods on the WDCM database are shown in Fig. 6. All methods have satisfying visualization results on the WDCM database. The error maps in Fig. 6 indicate that different methods have different reconstruction results for different types of land covers. The land-cover type of the green rectangle is the lake. The proposed SSRAN and compared methods achieve satisfying results in the green rectangle, due to the smooth spatial distribution of the lake. The land-cover type of the red rectangle is the artificial building. The proposed SSRAN performs better than PWC, J-SLoL, SSCNN, and SSJSRN since SSRAN can effectively exploit the spatial information and spectral signatures. However, the results of all methods are relatively inferior in the black rectangle. The pixels in the black rectangle are composed of different proportions of plants and soils. In practice, the diversity of the proportions of plants and soils increases the challenge of the MSISSR task.

2) On the Pavia University Database: The ground sample distance of the Pavia University database is 1.3 m, where a pixel also contains only a few endmembers or even one endmember. The compositions of spectral signatures on the Pavia University database are simple. Table V reports the quantitative results of SSRAN and compared methods on the Pavia University database. PWC still has poor performance on this database. SSCNN and SSJSRN show better performance than J-SLoL in terms of RMSE, SAM, UIQI, ERGAS, and PSNR since both spatial and spectral information are exploited in SSCNN and SSJSRN. Spa-Net performs better than Spe-Net in terms of RMSE, SAM, ERGAS, and PSNR, which indicates that the spatial information is important on the Pavia University database. Spa-Spe-Net outperforms Spe-Net and

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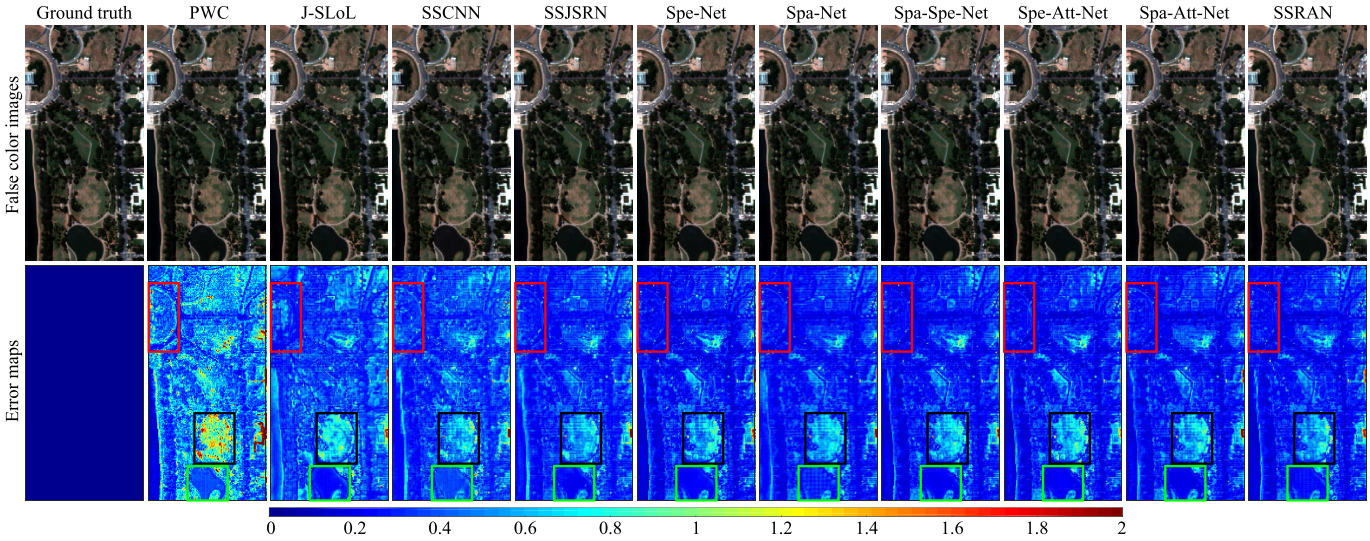


Fig. 6. False-color images and error maps of different methods on the WDCM database.

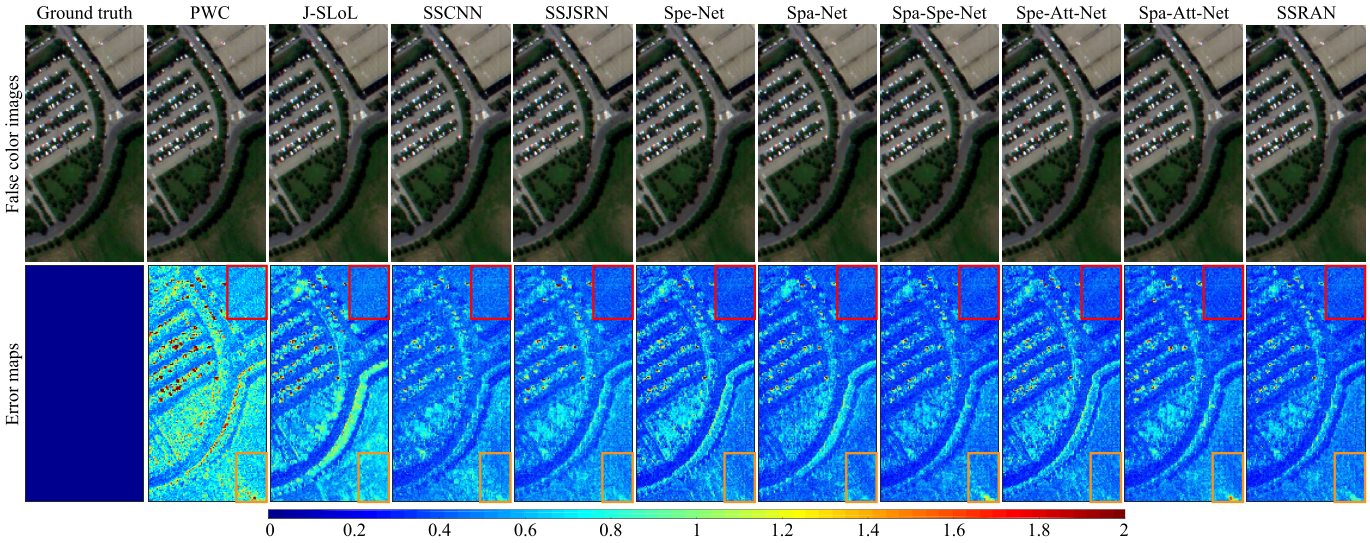


Fig. 7. False-color images and error maps of different methods on the Pavia University database.

Spa-Net in terms of RMSE, SAM, ERGAS, and PSNR, which demonstrates that spatial information and spectral signatures are complementary to MSISSR on the Pavia University database. Due to the well-designed spatial–spectral residual block and the neighboring spectral attention module, the proposed SSRAN outperforms compared methods in five evaluation metrics.

The visual results of the reconstructed HSIs and corresponding error maps on the Pavia University database are shown in Fig. 7. The proposed SSRAN and compared methods achieve favorable visual results on this database. As shown in Fig. 7, the reconstructed HSI mainly consists of the asphalt roads, meadows, trees, self-blocking bricks, and buildings. In the error maps, the land-cover type of the red rectangle is the artificial building. The proposed SSRAN shows superior performance in the red rectangle. The land-cover type of the orange rectangle is the meadow. The different denseness of the meadow in diverse locations results in different spectral signatures for the same land-cover type, which increases the difficulty of MSISSR. The proposed SSRAN still outperforms PWC, J-SLoL, SSCNN, and SSJSRN in the meadow.

3) On the Paris Database: MSIs captured by different imaging sensors usually have different ground sampling distances, which corresponds to different complexity of spectral signatures. The ground sample distance of the Paris database is 30 m, which may result in that a pixel in MSI and HSI contains dozens of endmembers. Too many endmembers in a pixel would make the compositions of spectral signatures very complicated. Complicated spectral signatures can make the spectral super-resolution challenging. Hence, learning the MSI-HSI mapping for the MSISSR task on the Paris database is more challenging than that on the WDCM and Pavia University databases. The quantitative results of SSRAN and compared methods on the Paris database are given in Table VI. Spa-Att-Net represents SSRAN without the spectral branch. As shown in Table VI, on the Paris

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TABLE VI
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE PARIS DATABASE

Methods	RMSE	SAM	UIQI	ERGAS	PSNR
PWC [9]	17.695	4.607	0.776	18.199	25.272
J-SLoL [9]	15.133	3.641	0.814	15.754	26.586
SSCNN [21]	11.926	3.045	0.891	12.335	28.640
SSJSRN [13]	10.923	3.079	0.905	11.342	29.387
Spe-Net	14.929	3.573	0.827	15.761	26.808
Spa-Net	10.943	3.003	0.906	11.428	29.342
Spa-Spe-Net	11.337	2.902	0.904	12.010	29.132
Spe-Att-Net	14.099	3.520	0.838	14.620	27.242
Spa-Att-Net	**10.511**	2.850	**0.914**	**10.950**	**29.755**
SSRAN	10.740	**2.848**	0.909	11.043	29.477

database, Spa-Att-Net achieves the best performance in terms of RMSE, UIQI, ERGAS, and PSNR. Comparing Spa-Att-Net with SSRAN in Table VI, it can be found that the spectral features from spectral signatures can reduce the experimental results of SSRAN. Although the performance of SSRAN on the Paris database is worse than that on the WDCM and Pavia University databases, the proposed SSRAN still outperforms compared methods.

As shown in Table VI, on the Paris database, PWC, J-SLoL, Spe-Net, and Spe-Att-Net that utilize only spectral signatures of MSI show poor performance. The poor performance of PWC, J-SLoL, Spe-Net, and Spe-Att-Net can indicate that the spectral signatures of MSI on the Paris databases are difficult to be exploited for the MSISSR task. Therefore, on the Paris databases, the spectral signatures of MSI are unimportant to the MSISSR task. SSCNN and SSJSRN achieve good results, which demonstrates that spatial information contributes to MSISSR on the Paris database. If both the spectral signatures and the spatial information are simultaneously exploited for the MSISSR task on the Paris database, the spectral signatures may disturb the experimental results. Comparing Spa-Net and Spa-Spe-Net, it can be found that the spectral branch in Spa-Spe-Net reduces the performance on the Paris database. However, this does not mean that the spectral branch in SSRAN should be removed. In practice, with the rapid development of remote sensing imaging technology, the ground sampling distances have been improved. There are few cases where the ground sampling distance of MSI is very low (e.g. 30 m). If the spectral branch in the proposed SSRAN is directly removed, the spectral signatures of MSI cannot be leveraged for the MSISSR task. As shown in Tables IV and V, removing the spectral branch can reduce the performance of SSRAN on the WDCM and Pavia University databases. As shown in Table VI, although adding the spectral branch can reduce the performance of SSRAN on the Paris database, the final result of SSRAN is still better than compared methods.

The visual results and corresponding error maps of SSRAN and compared methods on the Paris database are given in Fig. 8. As shown in Fig. 8, SSRAN and SSJSRN have clearer visual results than PWC and J-SLoL. The proposed SSRAN and compared methods achieve poor performance in the yellow rectangles. The possible reason for the poor performance of these methods in the yellow rectangles is that the numbers of pixels with different land-cover types in the training samples are unbalanced [49].

TABLE VII
RMSE AND SAM OF SSRAN WITH DIFFERENT NUMBERS OF TRAINING SAMPLES

Ratio	WDCM		Pavia University		Paris	
	RMSE	SAM	RMSE	SAM	RMSE	SAM
30%	0.918	1.081	2.411	2.417	10.906	3.142
40%	0.897	1.058	2.106	2.389	10.957	2.881
50%	**0.858**	0.946	2.154	2.341	10.740	2.848
60%	0.872	0.941	1.993	2.279	10.520	2.850
70%	0.861	**0.905**	**1.840**	**2.254**	**10.496**	**2.829**

TABLE VIII
RMSE AND SAM OF SSRAN WITH DIFFERENT SPATIAL SIZES

Spatial sizes	WDCM		Pavia University		Paris	
	RMSE	SAM	RMSE	SAM	RMSE	SAM
2×2	0.883	0.968	2.224	2.383	12.386	3.131
4×4	**0.858**	**0.946**	2.154	2.341	10.740	**2.848**
6×6	0.863	0.962	2.126	2.337	**10.540**	2.881
8×8	0.880	0.954	**2.101**	**2.336**	11.872	3.278

F. Effect of Number of Training Samples and the Spatial Size of Patches

1) Number of Training Samples: The number of training samples has a great impact on the performance of deep neural networks [13], [29]. In this section, the performance of the proposed SSRAN is explored under different numbers of training samples. The training sample ratio is set to 30%, 40%, 50%, 60%, and 70% to conduct experiments. RMSE and SAM of SSRAN are shown in Table VII. Deep networks usually require sufficient training samples for training [47]. As shown in Table VII, as the number of training samples increases, the performance of SSRAN is improved.

2) Spatial Size of Patches: In this section, experiments are conducted on the WDCM, Pavia University, and Paris databases to explore the effect of MSI patches with different spatial sizes on the performance of SSRAN. Different spatial sizes of MSI patches mean that different scopes of spatial information are utilized to reconstruct the HSI. The spatial size of MSI patches is set to 2×2, 4×4, 6×6, and 8×8. Experimental results are listed in Table VIII. As shown in Table VIII, the performance of SSRAN with 2×2 spatial size is worse than that with 4×4 spatial size since the smaller spatial size leads to the less spatial information utilized for MSISSR. On the WDCM and Paris databases, the performance of SSRAN with 8×8 spatial size is reduced. Large spatial size can also lead to the introduction of MSI pixels of other land-cover types, which reduces the quality of reconstructed HSI.

G. Time Cost

In this article, all methods are executed in a workstation, which is configured with Intel Core i7 5930k, 64 GB RAM, GeForce GTX TITAN X, and Ubuntu system. PWC and J-SLoL [9] are implemented in MATLAB R2014B and CPU. SSCNN, SSJSRN, Spe-Net, Spa-Net, Spa-Spe-Net,

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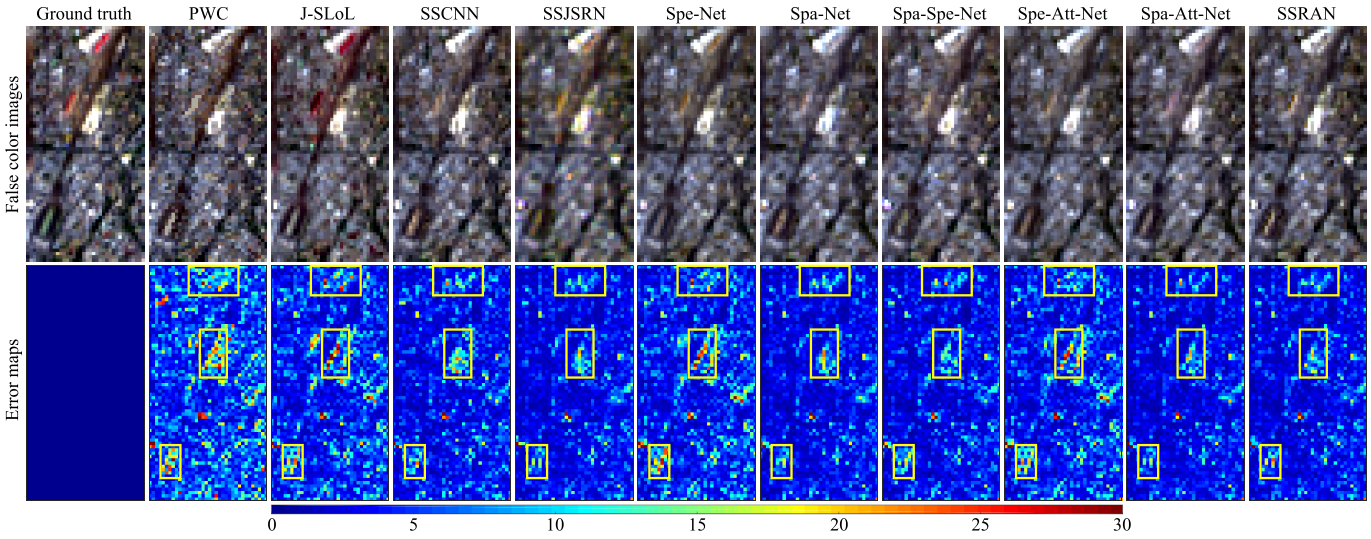


Fig. 8. False-color images and error maps of different methods on the Paris database.

TABLE IX
TRAINING TIME (MINUTES) AND TESTING TIME (SECONDS) OF DIFFERENT METHODS

Time cost	WDCM		Pavia University		Paris	
	Train	Test	Train	Test	Train	Test
PWC [9]	-	3.737	-	2.303	-	0.125
J-SLoL [9]	53.023	68.761	27.861	80.057	1.348	41.893
SSCNN [21]	28.651	0.316	19.860	0.134	2.101	0.017
SSJSRN [13]	167.148	1.072	62.574	0.395	9.323	0.064
Spe-Net	**5.253**	**0.063**	**3.876**	**0.037**	**0.418**	**0.002**
Spa-Net	6.280	0.083	4.483	0.039	0.494	0.003
Spa-Spe-Net	7.951	0.087	5.461	0.041	0.627	0.003
Spe-Att-Net	8.007	0.090	5.055	0.044	0.647	0.003
Spa-Att-Net	8.672	0.093	5.645	0.047	0.709	0.003
SSRAN	9.044	0.095	6.865	0.055	0.716	0.004

Spe-Att-Net, Spa-Att-Net, and SSRAN are executed in Tensorflow-1.4.1[8] and GPU. The training and testing time of all methods is listed in Table IX, where the batch size is 100 and the spatial size is 4×4. Since the PWC method does not need to be trained in advance, only the testing time is reported. In J-SLoL, the time to extract the HSI dictionary and MSI dictionary from the training set is counted as the training time. The time to extract the sparse coefficients of MSI and reconstruct the HSI is counted as the testing time. In SSJSRN, the 3-D convolution is utilized as the basic component to extract spatial–spectral features of MSI. Due to the low computational efficiency of 3-D convolution, the computational efficiency of SSJSRN is low. As shown in Table IX, the proposed SSRAN has higher computational efficiency than SSCNN and SSJSRN since the proposed SSRAN is composed of several 1-D and 2-D convolution layers.

H. Discussion

As shown in Tables IV–VI, the importance of the spatial information and spectral signatures in MSI is different from the MSISSR task on different databases. As shown in Tables IV and V, the performance of Spe-Net, Spa-Net, and Spa-Spe-Net demonstrates that both the spatial information and the spectral signatures of MSI is important for MSISSR on the WDCM and Pavia University databases. However, as reported in Table VI, the performance of Spe-Att-Net and Spa-Att-Net indicates that the spectral signatures of MSI are unimportant, and the spatial information of MSI is important for MSISSR on the Paris database. The proposed SSRAN cannot fully eliminate the negative effects of spectral signatures on the Paris database since there is no specially designed module to explicitly measure the importance of the spatial information and spectral signatures in SSRAN.

In future work, we will pay attention to the adaptive use of spectral signatures and spatial information in MSI for the MSISSR task. We will study new appropriate network structures, which can explicitly model the importance of the spectral signatures and the spatial information of MSI. We will also study new structures of the spectral branch to extract effective spectral features from the complicated spectral signatures of MSI. Then, these appropriate network structures can adaptively select and utilize the spectral features and the spatial information according to their importance.

V. CONCLUSION

In this article, an SSRAN is proposed to learn the MSI-to-HSI mapping for MSISSR. In the nonlinear mapping part of SSRAN, the spatial–spectral residual blocks are proposed to explore spatial and spectral information of MSI for obtaining the features of the HSI patch. In the reconstruction part of SSRAN, a neighboring spectral attention module is proposed to explicitly constrain the reconstructed HSI to maintain the correlation among neighboring spectral bands. Experiments on simulated and real databases demonstrate the performance of SSRAN exceeds that of the state-of-the-art methods. However, experimental results also show that the importance of spatial information and spectral signatures in MSI on different databases is different. According to the

[8]https://github.com/tensorflow/tensorflow

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characteristics of the database, adaptive fusion of spatial features and spectral features still needs further research. In future, we will study new spatial–spectral deep networks to adaptively aggregate spatial features and spectral features of MSI for reconstructing the HSI.

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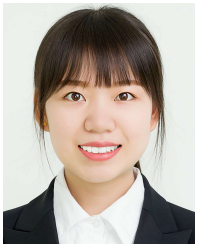
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